



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 01

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Program:

```
def linear_search(arr, key):  
    for i in range(len(arr)):  
        if arr[i] == key:  
            return i  
    return -1  
  
arr = [10, 25, 30, 45, 50]  
key = 30  
  
result = linear_search(arr, key)  
if result != -1:  
    print("Key found at index", result)  
else:  
    print("Key not found")
```

Output:

```
main.py Visualize Run ▶ Output  
1 def linear_search(arr, key):  
2     for i in range(len(arr)):  
3         if arr[i] == key:  
4             return i  
5     return -1  
6 arr = [10, 25, 30, 45, 50]  
7 key = 30  
8 result = linear_search(arr, key)  
9 if result != -1:  
10     print("Key found at index", result)  
11 else:  
12     print("Key not found")  
Key found at index 2
```